

Bridging the Digital Divide in Public Health: A Comparative Evaluation of Retrieval-Augmented Generation for Universal Health Coverage Intelligence in LMIC Nations

Good Reporting of A Mixed Methods Study (GRAMMS) Checklist: GRAMMS is selected as the primary reporting standard because this study employs an embedded mixed-methods design in which quantitative experimental findings (Phase 1) directly inform qualitative governance framework development (Phase 2).

GRAMMS Criterion	Focus Area	Remarks (Page No. – Para - description)
1. Justification for mixed methods	State the justification for utilizing mixed methods in this study.	1 / Abstract (para 3) & Section 4.1 : Mixed methods is used to combine quantitative platform performance with qualitative governance framework development
2. Description of design	Describe the type, priority and sequence of the different methods of study employed.	11 / Sec 4.0 to 4.3- Methods – Research Design ; Embedded quantitative factorial experiment (Phase 1) followed by qualitative framework development (Phase 2)
3. Methods detail	Describe the sampling approach for both quantitative and qualitative data collection and description of data analysis method for both methods.	12 -14 / Sec 4.2 - 4.6- Methods – Experimental Design; Purposive sampling; 80 WHO SDG documents, 200 query-response pairs; ANOVA + Fleiss' κ (quant); framework synthesis (qual)
4. Integration	State how the quantitative and qualitative data was integrated.	22 - 24 / Sec 5.3 - Results – Governance Interpretation: Quantitative failure modes mapped to RAG-PHI governance layers; integrated in Section 5.3 – 5.4 and Figure 1
5. Method limitations	Identify the limitations associated with combining the two methods used for the study (mixed methods).	26 – 28 / Sec 6.2 - Discussion : Limitations & FRDs: Limitation: WHO SDG dataset of 10 years is used for knowledge base; the results were analyzed using statistical analysis and experiment based query testing using docker terminal – there is no validation yet of the framework since it is conceptual based yet.
6. Insights from integration	Provide details of any new knowledge gained through the integration of quantitative and qualitative data.	27 -28 / Sec 6.1 – Discussion & Implications ; Integration revealed configuration choices are governance decisions in 6.1 – 6.3 and Table 4; RAG-PHI framework emerged from failure-mode analysis

Note: GRAMMS = O'Cathain et al. (2008). All six criteria are addressed within the main manuscript. The mixed-methods justification follows the WHO DHSPR 'Changing Mindsets' strategy, which advocates for combining 'fixed' quantitative methods with 'flexible' qualitative designs when investigating complex health system challenges

JARS - Mixed Methods Article Reporting Standards (MMARS):

The table below provides the complete JARS-Mixed checklist aligned with APA (2020) MMARS standards, augmented with seven manuscript-specific items (Items 4p–7a) appropriate to AI platform evaluation studies. Items marked 'Not applicable' are stated as such per JARS-Mixed guidance.

<i>Section /Topic</i>	<i>Item No</i>	<i>Checklist Item</i>	<i>Reported on Page Number/Line Number</i>	<i>Reported on Section/Paragraph</i>
Title	1a	Quant - Identify the populations or data sources studied.	Title page (Separately enclosed)	Title, Abstract
Title	1b	Mixed - Refrain from using words that are either purely qualitative (e.g., “explore,” “understand”) or purely quantitative (e.g., “determinants,” “correlates”) when a mixed methods framing is more accurate.	Title page	Title reflects mixed design
Title	1c	Mixed - Reference the mixed methods, qualitative/conceptual, and quantitative methods used.	Method	Abstract, Methodology – Pragmatic mixed-methods embedded design is used in the study

<i>Section /Topic</i>	<i>Item No</i>	<i>Checklist Item</i>	<i>Reported on Page Number/Line Number</i>	<i>Reported on Section/Paragraph</i>
Abstract	2a	Mixed - Indicate the mixed methods design, including types of participants or data sources, analytic strategy, main results/findings, and major implications/significance.	1 -2	Abstract
Introduction	3a	Quant - State the importance of the problem, including theoretical or practical implications.	3 - 4	Introduction
Introduction	3b	Qual - Review, critique, and synthesize the applicable literature to identify key issues, debates, and theoretical frameworks in the relevant literature to clarify barriers, knowledge gaps, or practical needs.	4 - 6	Literature Review 2.1-2.3
Introduction	3c	Mixed - State three types of research objectives/aims/goals: qualitative/conceptual, quantitative, and mixed methods. Order these goals to reflect the mixed methods design used.	4	Research Questions
Method	4a	Mixed - Explain why mixed methods research is appropriate as a methodology given the paper's goals.	10 - 14	Section 4.0 – 4.6

<i>Section /Topic</i>	<i>Item No</i>	<i>Checklist Item</i>	<i>Reported on Page Number/Line Number</i>	<i>Reported on Section/Paragraph</i>
Method	4b	Mixed - Indicate the qualitative/conceptual approach to inquiry and the quantitative design used within the mixed methods design type.	10 -12	Section 4.1 - 4.2(a)
Method	4c	Quant - Report inclusion and exclusion criteria for the quantitative component, including any restrictions relevant to documents, queries, platforms, models, or evaluation units.	11 -13	Section 4.2 - 4.3
Method	4d	Qual - Describe the participants/data source selection process and inclusion/exclusion criteria for the qualitative or conceptual component.	10 - 14	4.1 - 4.3: The qualitative framework development phase used purposive theoretical sampling, selecting failure-mode categories from the factorial experiment that represented distinct governance-layer violations.

<i>Section /Topic</i>	<i>Item No</i>	<i>Checklist Item</i>	<i>Reported on Page Number/Line Number</i>	<i>Reported on Section/Paragraph</i>
Method	4e	Quant - Describe settings and locations where data were collected or generated, as well as dates/time period of data collection.	Additional supplementary 22	Experimental evaluation was conducted in 2025 using local institutional hardware. [Institution blinded for review] ".
Method	4f	Qual - Provide the general context for the study, including when data were collected and the sites or sources of data collection.	Additional supplementary 3 - 7 & 22	Framework coding was conducted during 2025 using documented experimental outputs as primary data.
Method	4g	Quant - Define all primary and secondary measures, outcomes, and covariates.	11 -13	Section 4.2 - 4.3
Method	4h	Quant - Report major data-source characteristics and important topic-specific characteristics.	11 15 – Table 2 Additional Supplementary 3 - 4	Table 1: Methodology Summary (JARS-Mixed Items 4a–4r), 4.2 Experimental Design (JARS Items 4p–4q) in Table S1, S2 & Table 2: Comprehensive Performance Metrics by Experimental Condition

Section /Topic	Item No	Checklist Item	Reported on Page Number/Line Number	Reported on Section/Paragraph
Method	4i	Quant - Describe procedures for selecting data points, cases,. prompts, queries, documents, or participants	11 - 12	Section 4.2 – 4.2 (a)
Method	4j	Quant - Describe agreements, permissions, compensation, or payments, if applicable. If not applicable, state so.	-	Not applicable
Method	4k	Qual : State the form of data collected (e.g., interviews, documents, field notes, model outputs, observations, memos, coded text).	13	Section 4.4
Method	4l	Qual - Describe the questions, prompts, or units used in data collection: content of central questions and form of questions (e.g., open vs. closed).	11 - 12 33 3 – 4	Section 4.2 – 4.2 (a) + Appendix A + Additional supplementary
Method	4m	Qual - For interview, document, or text-based studies, indicate the duration, volume, or range of the data-collection process, where relevant.	12 – 13	Section 4.3
Method	4n	Qual - Describe any incentives or compensation, and provide assurance of relevant ethical processes, consent procedures, or state that these were not applicable.	13 -14	Section 4.5

Section /Topic	Item No	Checklist Item	Reported on Page Number/Line Number	Reported on Section/Paragraph
Method	4o	Mixed/General - Describe institutional review board approval, exemption, non-applicability, ethical standards met, and any safety or governance procedures.	13 – 14 28 - 30	Section 4.5 & Additional Information
Results	5a	Quant - Provide information detailing the statistical and data-analytic methods used and the corresponding quantitative findings.	14 - 22	Section 5.1-5.2; Tables 3 & 4
Results	5b	Qual - Describe qualitative/conceptual findings (e.g., themes, categories, narratives, constructs, framework elements) and the meanings derived from the analysis.	6 - 9	3.0 Theoretical and Conceptual Framework of RAG-PHI layers; Fig 1 as Tiff in GitHub repository
Results	5c	Qual - Demonstrate the analytic process of reaching findings (e.g., coding structure, excerpts, quotes, examples, matrices, or analytic trail).	22 – 24	Section 5.3, Governance Interpretation of Failure Modes and Integration of Findings (JARS Items 5b–5f): Table 4
Results	5d	Mixed - Indicate how the qualitative/conceptual and quantitative results were mixed or integrated.	22 - 24	Section 5.3

<i>Section /Topic</i>	<i>Item No</i>	<i>Checklist Item</i>	<i>Reported on Page Number/Line Number</i>	<i>Reported on Section/Paragraph</i>
Discussion	6a	Describe the central contributions and their significance in advancing disciplinary understanding.	24 -26	Section 6 – 6.3
Discussion	6b	Discuss similarities and differences between reported results and prior work.	25 -26	Section 6.2
Discussion	6c	Identify the study’s strengths and limitations.	25 - 26	Section 6.2

Optional manuscript-specific rows:

<i>Section /Topic</i>	<i>Item No</i>	<i>Checklist Item</i>	<i>Reported on Page Number/Line Number</i>	<i>Reported on Section /Paragraph</i>
Method	4p	Quant - Describe the experimental setup, including platforms, language models, parameter configurations, and retrieval settings.	11 -14 Additional supplement 5 - 21	Section 4.2 – 4.2 (a) Table S3 – S9
Method	4q	Quant - Define factual accuracy, hallucination rate, protocol adherence, and response consistency, including ground-truth procedures.	13	Section 4.4
Method	4r	Mixed - Describe the integration strategy linking quantitative platform evaluation to conceptual/theoretical framework development.	6 - 9, 22 - 24	Section 3.0 – 3.2; 5.3

<i>Section/ Topic</i>	<i>Item No</i>	<i>Checklist Item</i>	<i>Reported on Page Number/ Line Number</i>	<i>Reported on Section/ Paragraph</i>
Results	5e	Quant - Report comparative findings across platforms, models, and parameter conditions.	8 - 23	Tables 1 - 4
Results	5f	Mixed - Present the integrated findings that informed the RAG-PHI conceptual and theoretical framework.	22 – 24 , 8, 17-21 22 - 23	Section 5.3, Table 1, 3& 4
Discussion	6d	Mixed - Explain how integration of findings generated new insights for governance, equity, and public health intelligence.	24 -26	Section 6.1- 6.3
End matter	7a	Declaration - State the use of generative AI and AI-assisted technologies in the manuscript preparation process, if required by the journal.	28 - 30	Declaration of AI usage
All standard items (1a–6c) are fully addressed in the main manuscript				